PROJECT BRIEF · PYTHON (GR 6–8) INTERMEDIATE: PYTHON GAME CODERS · LEVEL 2

Lesson 4: Project: Turtle Art Studio

Module 1 | 60 minutes | functions, input, menus | Grades 7–11 · ages 12–16

Name: _____ Class: _____ Date: _____

Your mission
Build an interactive drawing studio where the user picks shapes, colours and sizes, and every shape has its own function.

Checklist

- [ ] At least 3 shape functions, each with a size parameter
- [ ] Asks for shape, colour and size with input()
- [ ] A while loop that keeps drawing until the user types done
- [ ] A friendly Sorry message for unknown shapes
- [ ] Shapes moved apart and counted, with a final message
- [ ] A title screen and a “Made by” comment

Milestones

1. Studio sketched on paper: shapes, menu questions and colours — done on: _____
2. Each shape function works when called on its own — done on: _____
3. Menu with if / elif / else draws the chosen shape — done on: _____
4. The loop runs until done; a friend has made a picture with it — done on: _____

Plan it

Write the steps of your program in plain English (pseudocode) before you code.

How it will be marked

Criterion	1 · Beginning	2 · Developing	3 · Secure	4 · Excellent

Program works	Crashes or draws nothing	Draws one shape, then stops or crashes	Draws every shape the user asks for and stops on done	Works perfectly and handles odd answers like STAR or banana
Functions	No functions	One function, or functions without parameters	3 or more shape functions with a size parameter	Extra helper functions (like move_to_spot) and no repeated code
User experience	No instructions for the user	Asks questions but gives no feedback	Clear menu, feedback after each shape and a final message	Fun title screen, shapes laid out neatly and a celebration at the end
Testing and code quality	Hard to read; never tested	Some comments; tested once	Each function tested; comments explain each part	Tested by another person and improved from their feedback

Want more?

- Add a “random” choice that picks a random shape, colour and size
- Let the user choose where each shape goes with x and y numbers
- Add a “spin” option that repeats the chosen shape 12 times like a rangoli

Two stars and a wish

Ask a friend to test your project.

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